

CROWN AND BRIDGES

PROVISIONAL RESTORATION (TEMPORARY RESTORATION)

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(TEMPORARY RESTORATION)

It is important because the prepared tooth or teeth must be protected and the patient kept comfortable while the final cast restoration is being fabricated.



Temporary restorations

They are defined as those restorations that are used during the time between tooth preparation and final placement of definitive (permanent) crown or bridge restorations.



A good provisional restoration should satisfy the following requirements:

1-Pulpal protection.

2-Positional stability.

3-Occlusal function.

4-Easily cleaned.

5-Non-impinging Margins.

6-Strength & retention.

7-Esthetic.

8- Solution for trial cases.

1-Pulpal protection:

When preparing the tooth, a layer of tooth enamel and a part of the dentin are removed, a large amount like what is done in a jacket crown, which may expose the tooth pulp to be closer to the surface of the tooth, making it subjected to allergy or damage from heat, cold, acidity, or sweetness of foodstuffs.

Therefore, the tooth should be covered with a substance that has no effect on the tooth and fixed it with **Zink Oxide Eugenol**, which has an analgesic and treating effect on the tooth pulp.

2-Positional stability:

During preparation a contact is opened and an occlusal clearance which will lead to overeruption of both the abutment and opposing teeth and tilting of adjacent teeth. Putting temporary restoration will fix teeth in position and it will support the lip and cheek from collapse after tooth extraction.

A temporary restoration will prevent the gingiva from overgrowth after tooth preparation especially in interproximal area.

3-Occlusal function:

The restoration will restore the shape of occlusal surfaces of the teeth in addition to substitution of missing teeth which will improve the function of the mouth.

4-Easily cleaned:

The restoration must be done in a design which makes using it without accumulation of food around the abutment tooth by selecting a proper design of pontics (like bullet pontic).

5-Non-impinging Margins:

The margins of restoration must fit on the finishing lines without over-extension which may press on gingival margin.

6-Strength & retention:

The restoration must be made from strong material that must withstand the forces of mastication like metal or polycarbonate and must be designed to be retentive on abutments.

7-Esthetic:

The restoration must give the patient good appearance and color by adding the missing the missing teeth on it.

8-Trial cases:

In cases which require special design like big space of a tooth and certain size of a tooth must be tried to select the proper size.

Types of provisional crowns:

Ready made crowns:

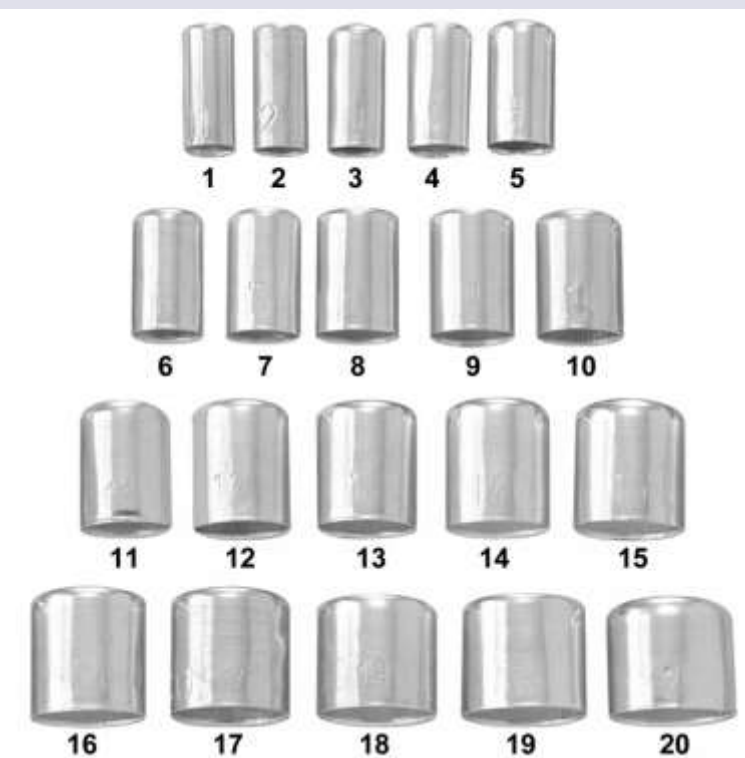
Metal crowns:

Mainly used for posterior teeth.

They are made from stainless steel, nickel chromium or aluminum & the last one is the most useful because it is soft.

The metal crowns are of 2 types:

- 1) Flat topped cylindrical types
- 2) Morphologically contoured type.



Disadvantages include:

1- Unpleasant taste.

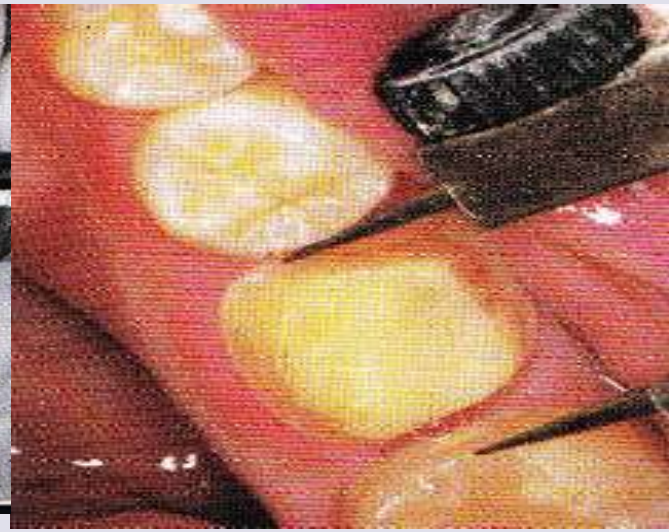
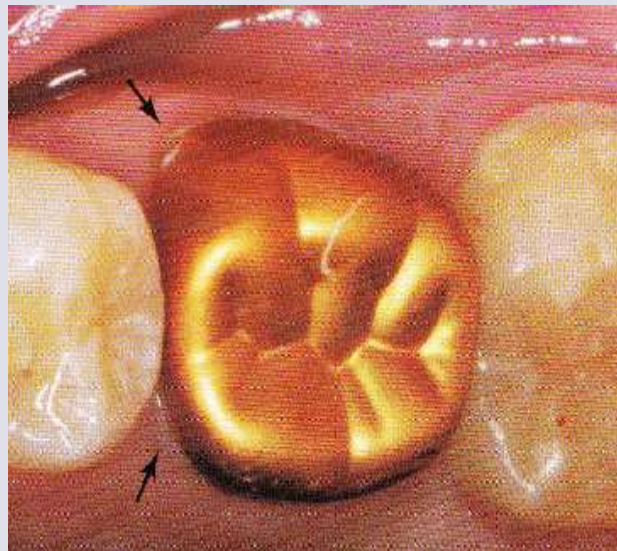
2- Poor esthetic.

3- Rapid wear of the crown.



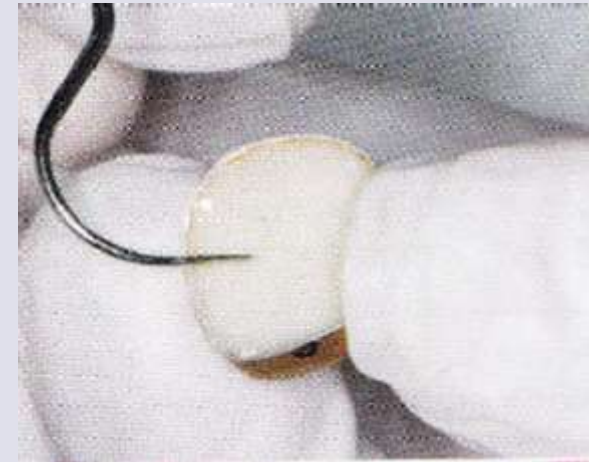
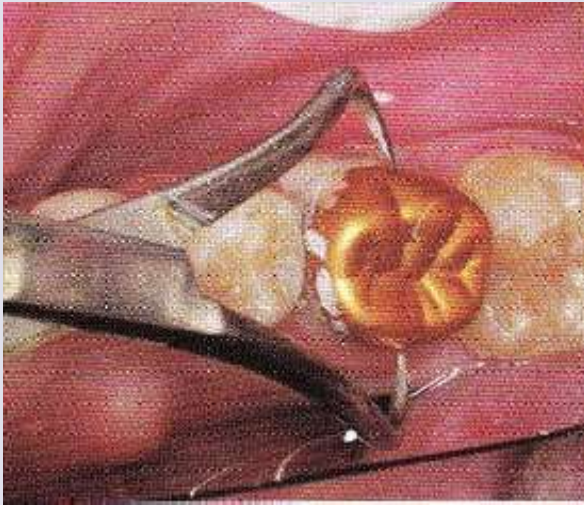
Clinical procedure:

- 1- Select the proper size & shape of the provisional crown.
- 2- Trim the margin to be fit to the finishing line and to accommodate the vertical height of the prepared tooth.
- 3- Seat the crown over the tooth and check the margins and occlusion.



4-Mix the resin, and fill the shell and place the shell over the tooth and remove the excess around the margins.

5- After 5 minutes, mark the margins and trim away any internal excess.



- 6-Reseat the crown and adjust the occlusion.
- 7- Further contouring to improve gingival embrasure.
- 8-Polish, clean, and cement the restoration.



Celluloid crown forms:

It is mainly used for anterior teeth, but can be used for posterior teeth also.

It acts as a mould for construction of provisional restoration.

It is filled with composite or tooth-colored resin to form the required crown.

Celluloid crowns



Clinical procedure:

- 1- Select the proper size and shape.
- 2- Try it on the tooth after trimming the margins.
- 3- Fill the crown with composite (cure with light).
- 4- Remove after setting, peel the crown, trim, check occlusion, smooth and polish then cement.



Polycarbonate crown forms

It is similar to acrylic crowns but made from polycarbonate polymeric material.

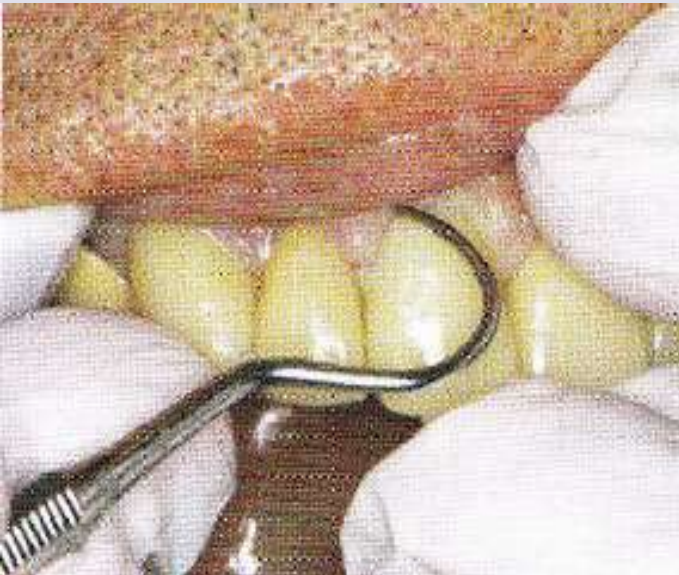
it is stronger and better than acrylic.

It is available in different sizes, shapes and shades.



Clinical procedure:

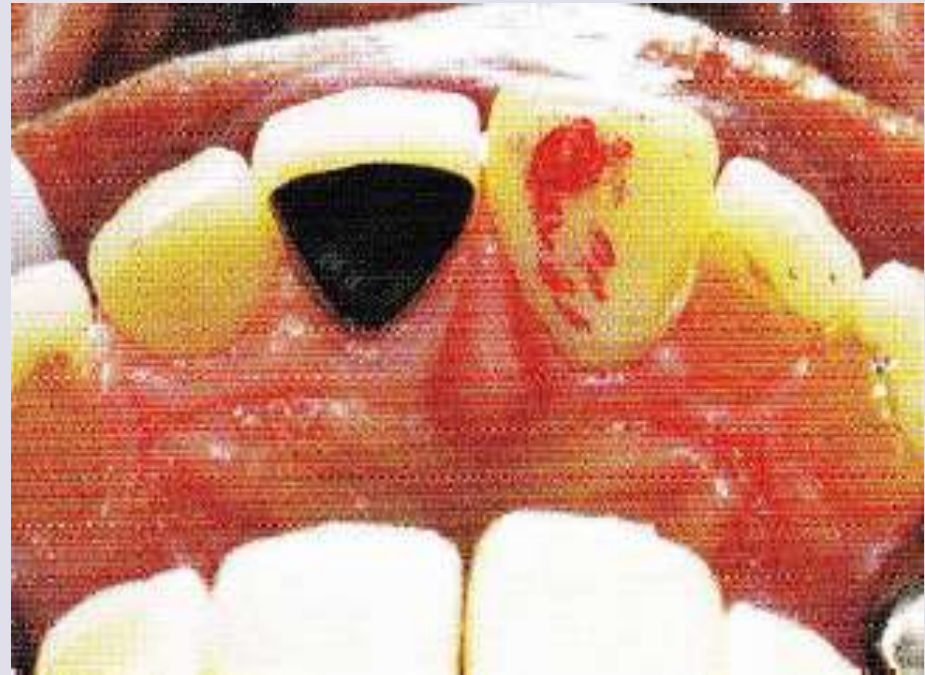
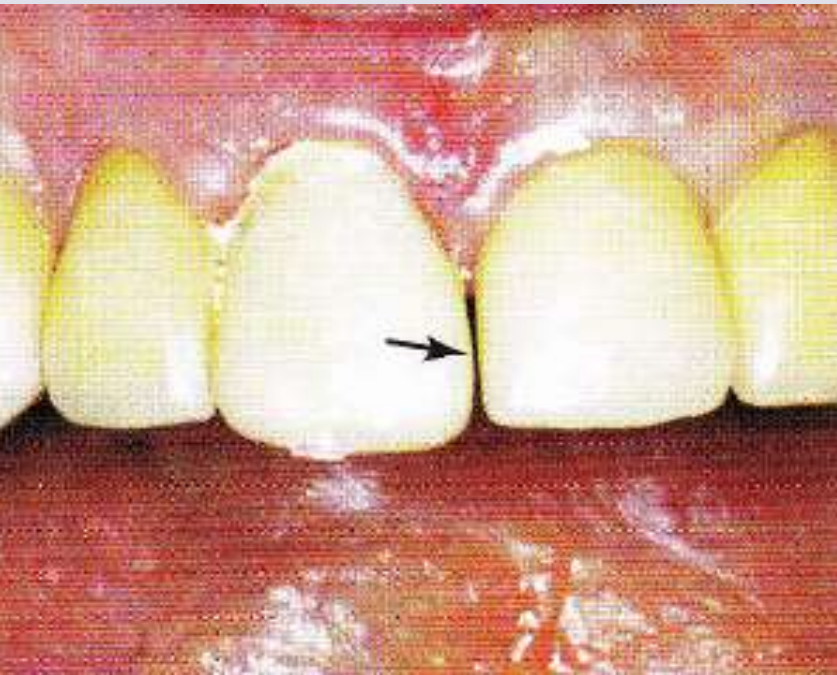
- 1- Measure the width and length.
- 2- Try the shell on the prepared tooth.
- 3- Mix the resin and fill the shell.
- 4- Remove excess and wait for setting.



Clinical procedure

5- Try- the crown, check high spots.

6- Remove the excess and cement the crown.



Acrylic ready-made crown:

It is used for anterior and posterior teeth.
It is supplied with different sizes & shades.

After choosing the proper size & shade the procedure is similar to that of polycarbonate crowns.



custom made restorations (crowns):

There are two methods: Direct and indirect

Direct impression method:

- 1- A preoperative impression with alginate or silicon rubber base is made and stored until preparation of teeth is completed.
- 2- Coat the tooth with separation medium.



3- Mix the resin or use the ready-packed cartridge.



4- load the tray with mixed resin(on the position of the prepared tooth only) and insert it inside the patient mouth.



5- After complete polymerization (if the reaction is exothermic you have to take it out from the patient mouth before that), the formed crown is removed and any excess material is trimmed and then reseal and check occlusion and margins, then cement crown by temporary cement.



Indirect method:

To prevent tooth irritation we can take two impressions: one preoperative and one postoperative impression. Pour postoperative one with quick set stone.

Remove the cast and repeat the same steps of direct method on the cast after putting separating medium.

Advantages:

- 1-No heat and monomer irritation to the tooth.**
- 2- Better margin.**
- 3- less chair time for the patient.**

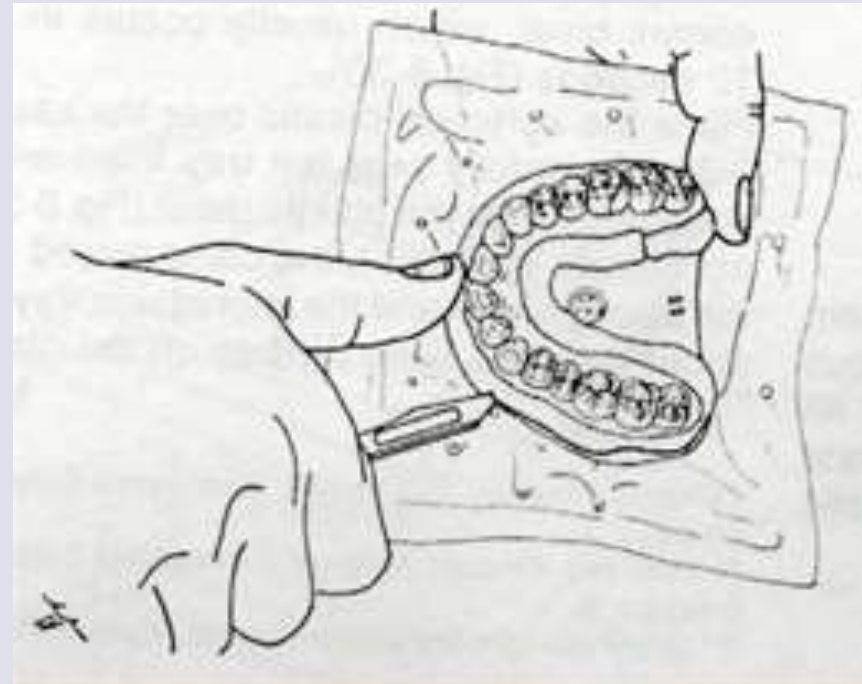
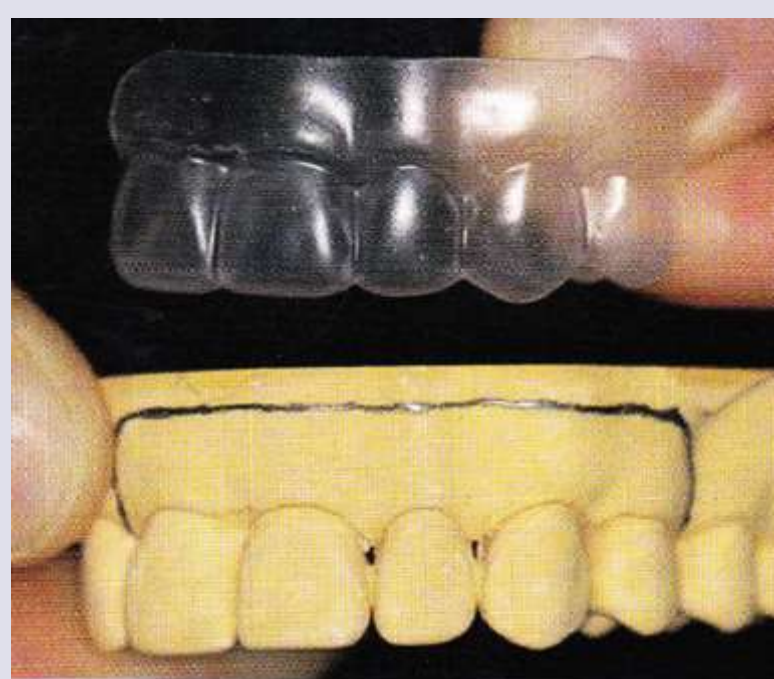
Custom provisional bridge techniques:

it is important to take primary alginate impression to produce study cast which is very useful during the designing of temporary restorations.



Indirect technique:

Adapt denture teeth on study cast in area of missing teeth and produce a frame for the bridge area.



1- Indirect technique:

(Dental BIOSTAR machine):
It is used to prepare a frame
(mold) for custom-made temporary
bridges or crowns.



Indirect technique:

Fill the frame with acrylic resin and insert it on a cast of prepared teeth.

Remove the set bridge, finish, polish and cement.



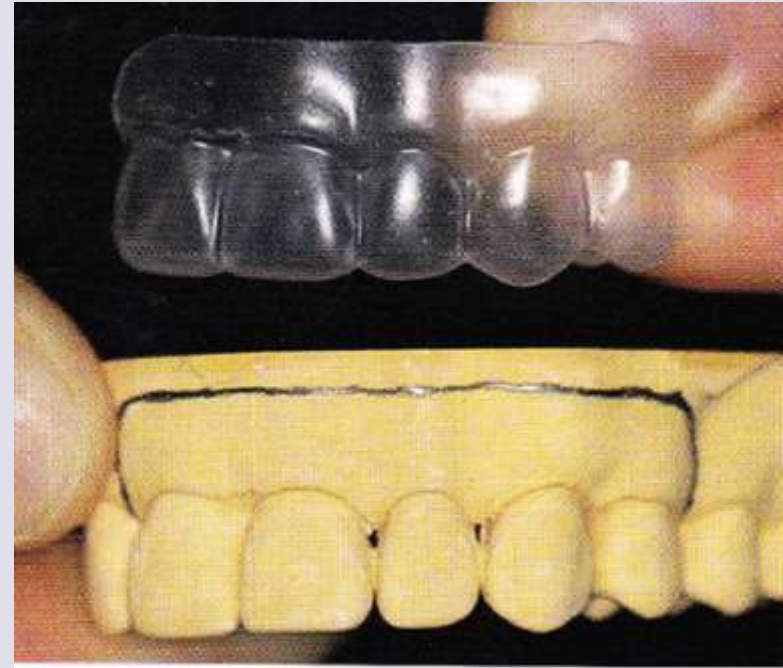
2- Direct temporary bridge technique:

- 1- Take elastomeric impression for study cast using heavy body before preparation and after adding the missing teeth.
- 2- Fill the area of bridge with resin and insert in patient's mouth after teeth preparation.
- 3- after setting remove polish and cement.



Another direct temporary bridge technique:

- 1- Prepare the celluloid transparent frame for bridge area by Biostar machine.
- 2- Fill frame with light-cured composite material and insert it on prepared teeth in patient's mouth.
- 3- Cure with light, remove from the mouth and then finish and cement.



3- Preprepared bridge technique:

1- Prepare abutment teeth on study cast.

2- Doing complete fabrication for the bridge **from tooth-colored hot-cured acrylic.**

3- after preparation of teeth seat the bridge after modification and cement it.



Temporary post crown:

Preformed:

ready-made provisional crown is made from tooth-colored acrylic with metal post inside. So we can trim excess and adapt them (by relining) to be seated perfectly on the prepared tooth.

Provisional post crown technique:

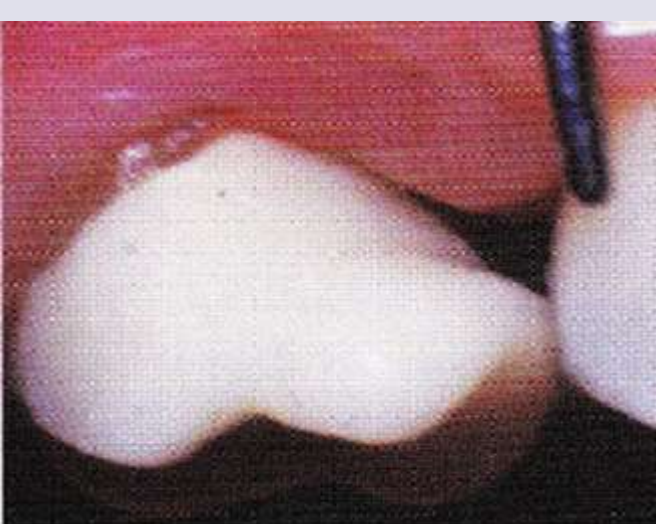
Custom made:

This can be fabricated by using standard temporary acrylic or celluloid crown by placing a piece of wire into the canal having a bend or serrations for retention.

Place the wire inside the canal

Seat the resin loaded (acrylic or celluloid) crown over all.

Remove after setting then finish, polish and cement.



Provisional post crown technique:

NOTE:

commonly, in cases of post crown the coronal portion is built up(as core) using ready-made post with composite or amalgam material then covered with the mentioned types of temporary crowns or bridges.





Thank You!

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